

Lab 4

Switch and Break Statements in C++

Learning Objectives

After completing this lab, after completing this lab we learnt:

- To understand the working of the switch statement in C++.
- To learn how to use the break statement inside switch cases.
- To implement decision-making programs using switch-case.
- To replace multiple if-else conditions using switch structure.

Switch Statement

The switch statement is a control structure in C++ used for decision making. It allows the execution of one block of code among many alternatives based on the value of a variable.

The break statement is used to terminate a case inside the switch. Without break, the program continues execution into the next cases (called fall-through).

Syntax:

```
switch(expression) { //Condition should be true , then cases runs
    case value1:
        // code block
        break;
    case value2:
        // code block
        break;
    default:
        // code block
}
```

Explanation:

- switch(expression) → checks variable value
- case → defines different conditions
- break → stops execution of further cases
- default → executes when no case matches

Example: Program Example (Switch Case with Break). Simple Calculator using Switch

```
int main() {  
    int a, b, choice;      // Variables for numbers and user choice  
    cout << "Enter first number: "; // Input first number  
    cin >> a;  
    cout << "Enter second number: "; // Input second number  
    cin >> b;  
    cout << "Select operation: ";    // Show menu  
    cout << "\n1. Addition";  
    cout << "\n2. Subtraction";  
    cout << "\n3. Multiplication";  
    cout << "\nEnter choice: ";  
    cin >> choice;           // Read user choice  
    switch(choice) {         // Start switch statement  
        case 1:              // Addition case  
            cout << "Sum = " << a + b << endl; // Add numbers  
            break;           // Exit switch  
        case 2:              // Subtraction case  
            cout << "Difference = " << a - b << endl;  
            break;           // Exit switch  
        case 3:              // Multiplication case  
            cout << "Product = " << a * b << endl;  
            break;           // Exit switch  
        default:             // Invalid input case  
            cout << "Invalid choice!" ; return 0; // End program  
    }  
}
```

Input:

Enter first number: 10

Enter second number: 5

Select operation: 1

Output:

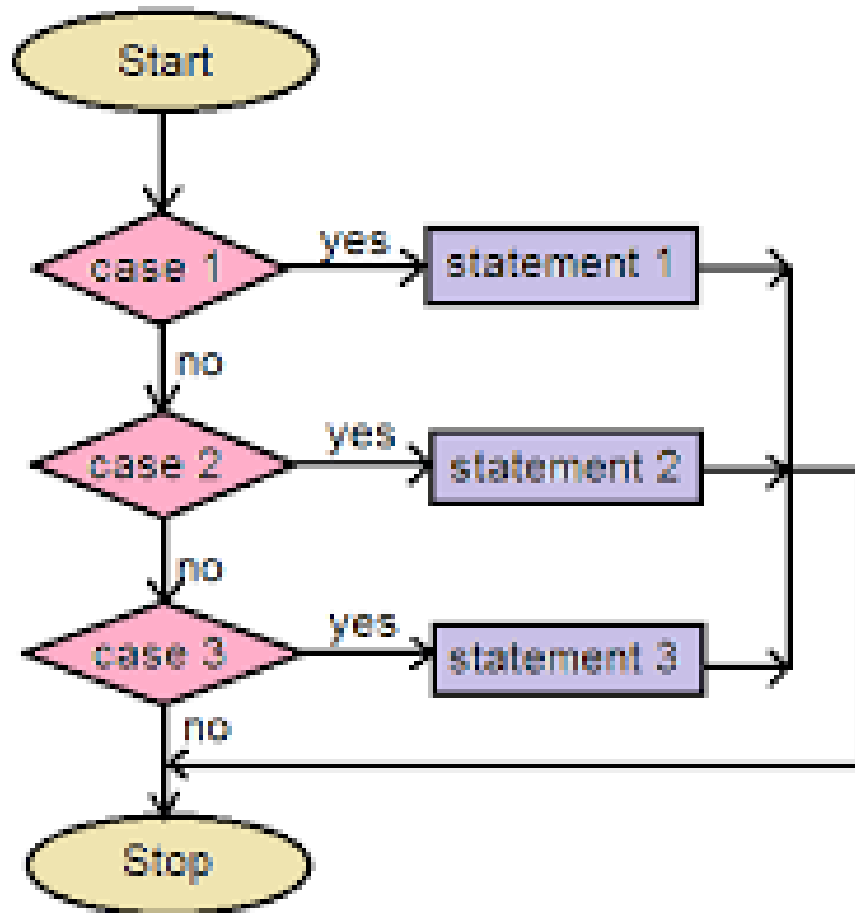
Sum = 15

Example: Day of Week using Switch

```
#include <iostream>    // Input/output library
using namespace std;   // Standard namespace
int main() {
    int day;           // Variable for day number
    cout << "Enter day number (1-7): ";
    cin >> day;
    switch(day) {      // Switch on day value
        case 1:
            cout << "Monday" << endl;
            break;
        case 2:
            cout << "Tuesday" << endl;
            break;
        case 3:
            cout << "Wednesday" << endl;
            break;
        case 4:
```

```
        cout << "Thursday" << endl;
        break;
    case 5:
        cout << "Friday" << endl;
        break;
    case 6:
        cout << "Saturday" << endl;
        break;
    case 7:
        cout << "Sunday" << endl;
        break;
    default:
        cout << "Invalid day number!" << endl;
    }
    return 0;
}
```

Flow Chart:



Continue Statement in C++

The continue statement is used inside loops to skip the current iteration and move directly to the next iteration of the loop.

It does not terminate the loop like break; instead, it only skips one step and continues execution.

Syntax:

```
continue;
```

When continue is executed inside a loop:

- Remaining code in the current iteration is skipped
- Control jumps to the next loop iteration

Examples:

```
#include <iostream>           // Input/output library
using namespace std;         // Standard namespace
int main() {
    for (int i = 1; i <= 5; i++) { // Loop from 1 to 5
        if (i == 3) {           // If i is 3
            continue;           // Skip this iteration
        }
        cout << i << endl;      // Print value
    }
    return 0;                 // End program
}
```

Output:

1
2
3
4

Comparison of Break and Continues statement:

Break	Continue
Break is used to terminate the execution of the loop.	Continue is not used to terminate the execution of loop.
It breaks the iteration.	It skips the iteration.
When this statement is executed, control will come out from the loop and executes the statement immediate after loop.	When this statement is executed, it will not come out of the loop but moves/jumps to the next iteration of loop.
Break is used with loops as well as switch case.	Continue is only used in loops, it is not used in switch case.

Lab5: Lab Tasks

Task 1: Write a C++ program that takes an input of a day's number and uses a switch statement to output the corresponding day's name.

```
#include <iostream>           // Header file

using namespace std;         // To use cin & cout without std::

int main() {                 // Main function starts

    int day;                  // Variable to store day number

    cout << "Enter day number (1-7): "; // Ask user for input

    cin >> day;                // Take input from user

    switch(day) {             // Switch checks entered day number

        case 1:                // If user enters 1

            cout << "Monday";  // Display Monday
```

```
break;                                // Exit switch

case 2:                               // If user enters 2

    cout << "Tuesday";               // Display Tuesday

    break;                            // Exit switch

case 3:                               // If user enters 3

    cout << "Wednesday";            // Display Wednesday

    break;                            // Exit switch

case 4:                               // If user enters 4

    cout << "Thursday";              // Display Thursday

    break;                            // Exit switch

case 5:                               // If user enters 5

    cout << "Friday";                // Display Friday

    break;                            // Exit switch

case 6:                               // If user enters 6

    cout << "Saturday";              // Display Saturday

    break;

case 7:                               // If user enters 7

    cout << "Sunday";                // Display Sunday

    break;                            // Exit switch

default:                             // If user enters wrong number

    cout << "Invalid day number";    // Show error

return 0;                             // Program ends
```

Task 2: Grade Evaluation Using Switch Statement

```
#include <iostream> // Header file

using namespace std; // Standard namespace

int main() { // Main function

    char grade; // Variable to store grade

    cout << "Enter your grade (A/B/C/D/F): "; // Ask user
    cin >> grade; // Take grade input

    switch(grade){ // Check entered grade

        case 'A': // If grade is A
            cout << "Excellent"; // Output message
            break; // Exit switch

        case 'B': // If grade is B
            cout << "Very Good"; // Output message
            break; // Exit switch

        case 'C': // If grade is C
            cout << "Good"; // Output message
            break; // Exit switch

        case 'F': // If grade is F
            cout << "Fail"; // Output message
            break; // Exit switch

        default: // Invalid input
            cout << "Invalid Grade"; // Show error
    }

    return 0; } // End program
```

Task 3: Unit Conversion Menu Using Switch Statement

```
#include <iostream>    // Header file

int main() {           // Main function

    int choice;        // Variable for menu choice

    float value;       // Variable for entered value

    float result;      // Variable for converted result

    cout << "Conversion Menu" << endl;    // Display heading

    cout << "1. Kilometers to Meters" << endl; // Option 1

    cout << "2. Meters to Centimeters" << endl; // Option 2

    cout << "3. Kilograms to Grams" << endl;    // Option 3

    cout << "4. Celsius to Fahrenheit" << endl; // Option 4

    cout << "Enter your choice: ";    // Ask user for choice

    cin >> choice;                    // Take choice input

    cout << "Enter value: ";          // Ask for value

    cin >> value;                      // Take value input

    switch(choice) {                  // Check selected option

        case 1:                      // Kilometers to Meters

            result = value * 1000;    // Conversion formula

            cout << "Result = " << result << " meters"; // Display

            break;                    // Exit switch
```

```

case 2:                                // Meters to Centimeters

    result = value * 100;              // Conversion formula

    cout << "Result = " << result << " centimeters"; // Display

    break;                             // Exit switch

case 3:                                // Kilograms to Grams

    result = value * 1000;             // Conversion formula

    cout << "Result = " << result << " grams"; // Display

    break;                             // Exit switch

case 4:                                // Celsius to Fahrenheit

    result = (value * 9 / 5) + 32;     // Conversion formula

    cout << "Result = " << result << " Fahrenheit"; // Display

    break;                             // Exit switch

default:                               // Invalid menu option

    cout << "Invalid choice"; // Show error

}

return 0; }                            // Program ends

```